

# **Secure Enterprise Network Design Using Cisco Packet Tracer**

## **1. Abstract**

This paper presents the design and implementation of a secure enterprise network using Cisco Packet Tracer. The project focuses on logical network segmentation through Virtual Local Area Networks (VLANs), inter-VLAN communication using Router-on-a-Stick configuration, dynamic IP address allocation using the Dynamic Host Configuration Protocol (DHCP), and traffic control through Extended Access Control Lists (ACLs). Three departments - Admin, Human Resources (HR), and IT are simulated to represent a real-world organizational environment. Security policies are enforced to restrict unauthorized access between sensitive departments while allowing required communication. Detailed testing and verification procedures are performed to validate both connectivity and security requirements. Troubleshooting is conducted to resolve protocol-level issues related to ICMP traffic and ACL behavior, demonstrating practical problem-solving skills. The results confirm that the implemented network meets the intended objectives of segmentation, security, and scalability. This independent work highlights hands-on experience in enterprise networking concepts and provides practical exposure relevant to advanced networking and information security domains.

## **2. Tools Used**

2.1 Cisco Packet Tracer 8.2.2

2.2 Cisco 2960 Switch

2.3 Cisco 2911 Router

2.4 PCs

### **3. Secure Enterprise Network Design Using Cisco Packet Tracer**

The increasing demand for secure and scalable enterprise networks necessitates effective traffic segmentation and access control mechanisms. Logical isolation of departments helps reduce broadcast traffic and enhances security. This paper demonstrates the practical implementation of these concepts using Cisco Packet Tracer.

## **4. Network Architecture**

### **4.1 VLAN Configuration**

The network is divided into three VLANs representing organizational departments. Each VLAN operates on a separate IP subnet to ensure logical isolation.

### **4.2 Inter-VLAN Routing**

Router-on-a-Stick is used to enable communication between VLANs. Sub-interfaces are configured on the router with IEEE 802.1Q encapsulation.

## **5. Security Implementation**

### **5.1 Extended Access Control Lists**

Extended ACLs are applied at the router interface connected to the HR VLAN. The ACL restricts HR hosts from accessing the Admin network while allowing return traffic initiated by Admin systems. This ensures confidentiality without disrupting legitimate communication.

## **6. Testing and Verification**

Connectivity testing is performed using ICMP ping between hosts within the same VLAN and across VLANs. Security verification confirms that access restrictions function as intended. Router and switch verification commands are used to validate configuration accuracy.

## 7. Results and Discussion

The implemented network successfully achieves departmental isolation, secure inter-VLAN routing, and controlled access. Troubleshooting of ICMP-related ACL behavior provided deeper insight into protocol-aware security configuration.

## 8. Conclusion

This mini project demonstrates a secure enterprise network design using industry-standard Cisco networking concepts. The implementation reflects real-world practices and strengthens practical understanding of network security and administration.

## 9. References

Cisco Networking Academy. (2023). *Introduction to networks*. Cisco Systems.

## 10. Tables

### VLAN and IP Addressing Scheme

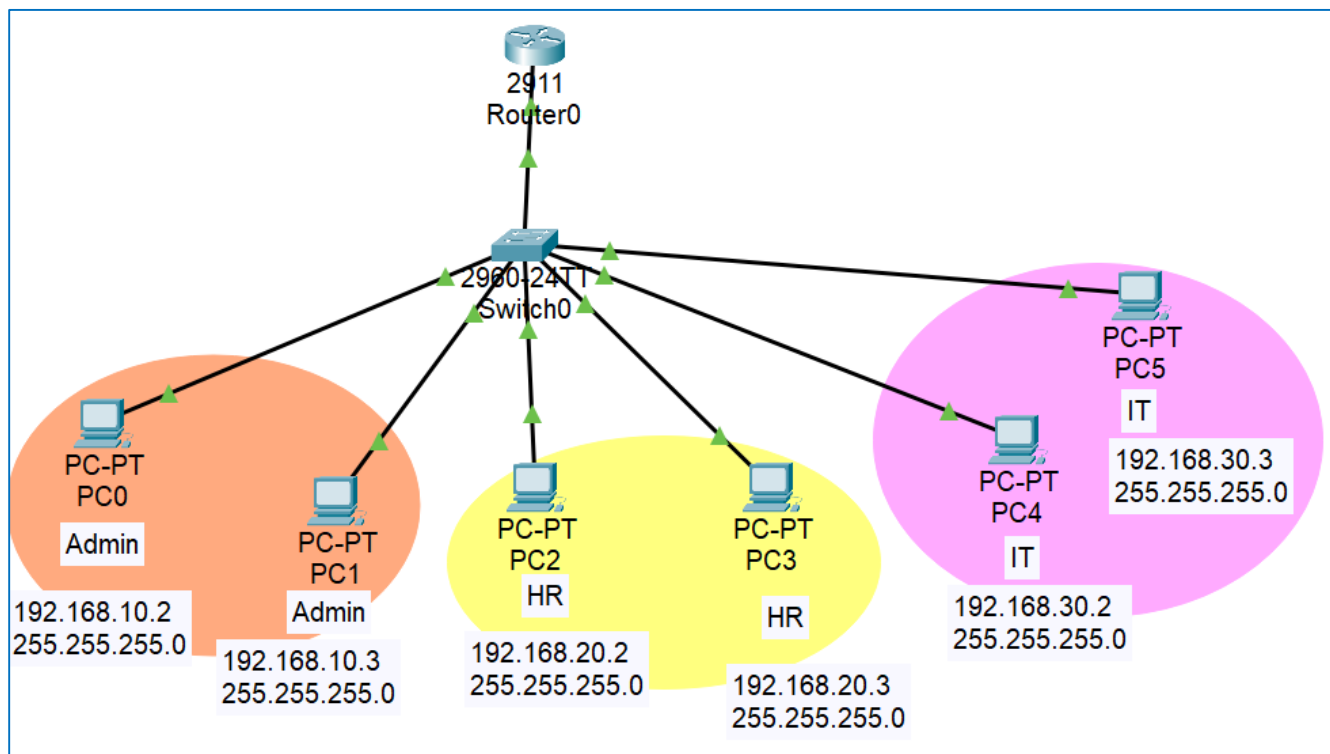
| VLAN ID | Department | Network Address | Default Gateway |
|---------|------------|-----------------|-----------------|
| 10      | Admin      | 192.168.10.0/24 | 192.168.10.1    |
| 20      | HR         | 192.168.20.0/24 | 192.168.20.1    |
| 30      | IT         | 192.168.30.0/24 | 192.168.30.1    |

| Source (PC) | Destination (PC) | VLANs | Expected Result | Actual Result | ✓/✗ |
|-------------|------------------|-------|-----------------|---------------|-----|
| PC2 (HR)    | PC0 (Admin)      | 20→10 | ✗ Blocked       | ✗ Unreachable | ✓   |
| PC0 (Admin) | PC2 (HR)         | 10→20 | ✓ Allowed       | ✓ Reply       | ✓   |
| PC2 (HR)    | PC4 (IT)         | 20→30 | ✓ Allowed       | ✓ Reply       | ✓   |

|             |          |       |           |         |   |
|-------------|----------|-------|-----------|---------|---|
| PC1 (Admin) | PC3 (HR) | 10→20 | ✓ Allowed | ✓ Reply | ✓ |
|-------------|----------|-------|-----------|---------|---|

This table summarizes the logical segmentation used in the network design.

## 11. Screenshots



*Fig 1: Network Topology*

```

C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
  
```

*Fig 2: Ping request from PC2 (HR) to PC0 (Admin)*

```

C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

*Fig 3: Ping request from PC2 (HR) to PC4(IT)*

```

C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=20ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 20ms, Average = 5ms

C:\>ping 192.168.30.3

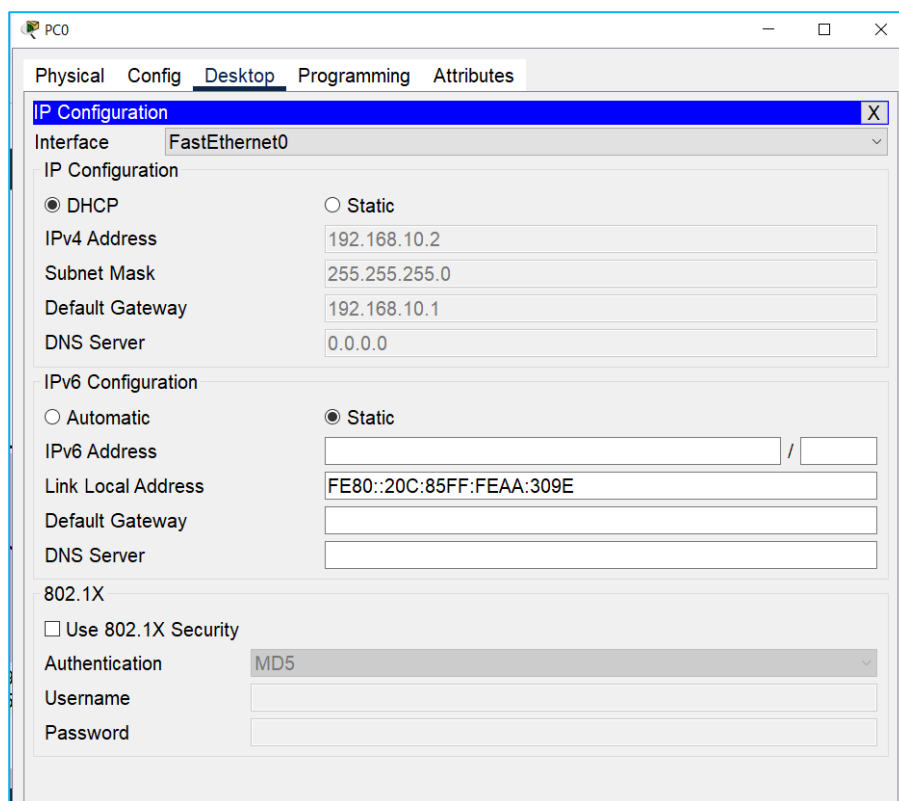
Pinging 192.168.30.3 with 32 bytes of data:

Reply from 192.168.30.3: bytes=32 time=1ms TTL=127
Reply from 192.168.30.3: bytes=32 time<1ms TTL=127
Reply from 192.168.30.3: bytes=32 time<1ms TTL=127
Reply from 192.168.30.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

```

*Fig 4: Ping request from PC0 (Admin) to PC2 (HR) & PC5 (IT)*



**Fig 5: DHCP assigning IPs to all the PCs**

```
Switch>show vlan brief
```

| VLAN | Name               | Status | Ports                                                                                                                                                        |
|------|--------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | default            | active | Fa0/8, Fa0/9, Fa0/10, Fa0/11<br>Fa0/12, Fa0/13, Fa0/14, Fa0/15<br>Fa0/16, Fa0/17, Fa0/18, Fa0/19<br>Fa0/20, Fa0/21, Fa0/22, Fa0/23<br>Fa0/24, Gig0/1, Gig0/2 |
| 10   | ADMIN              | active | Fa0/2, Fa0/3                                                                                                                                                 |
| 20   | HR                 | active | Fa0/4, Fa0/5                                                                                                                                                 |
| 30   | IT                 | active | Fa0/6, Fa0/7                                                                                                                                                 |
| 1002 | fddi-default       | active |                                                                                                                                                              |
| 1003 | token-ring-default | active |                                                                                                                                                              |
| 1004 | fddinet-default    | active |                                                                                                                                                              |
| 1005 | trnet-default      | active |                                                                                                                                                              |

```
Router(config)#access-list 100 permit icmp 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255 echo-reply
Router(config)#access-list 100 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
Router(config)#access-list 100 permit ip any any
Router(config)#interface g0/0.20
Router(config-subif)#ip access-group 100 in
Router(config-subif)#exit
```

```
Router#show access-lists
Extended IP access list 100
 10 permit icmp 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255 echo-reply (12 match(es))
 20 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255 (12 match(es))
 30 permit ip any any (12 match(es))
```

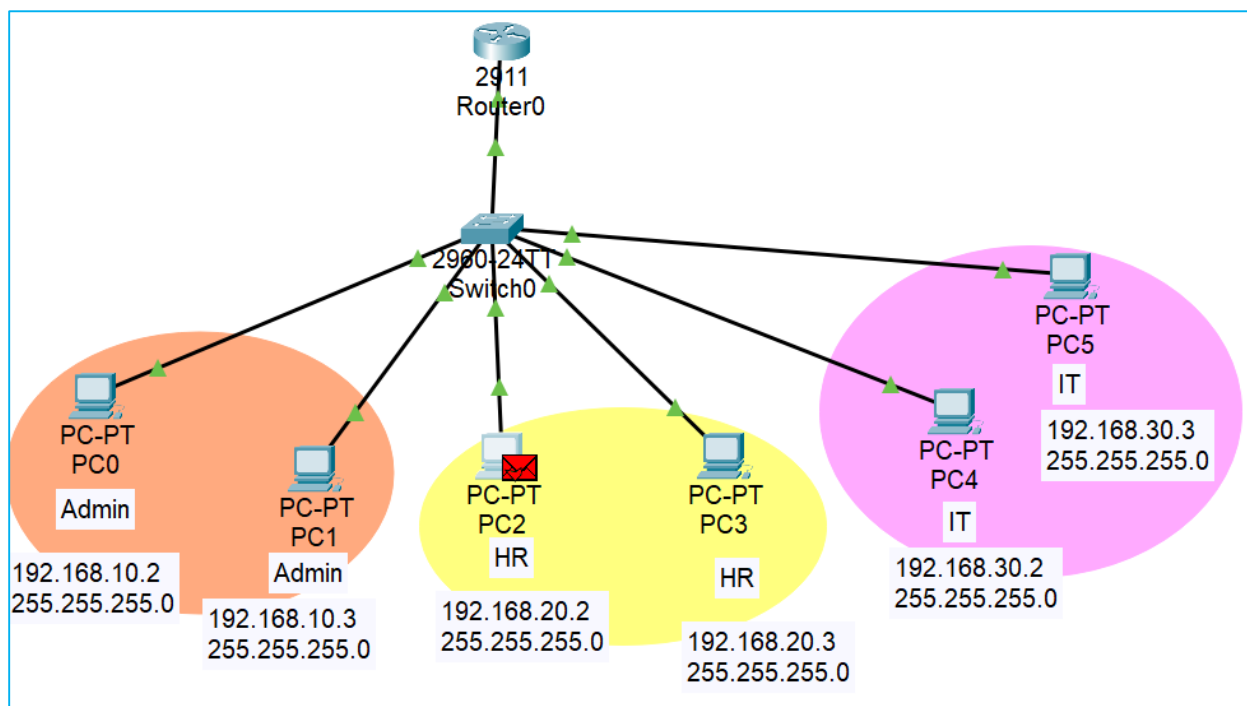
PDU Information at Device: PC2

**OSI Model**   Inbound PDU Details

At Device: PC2  
Source: Router0  
Destination: PC2

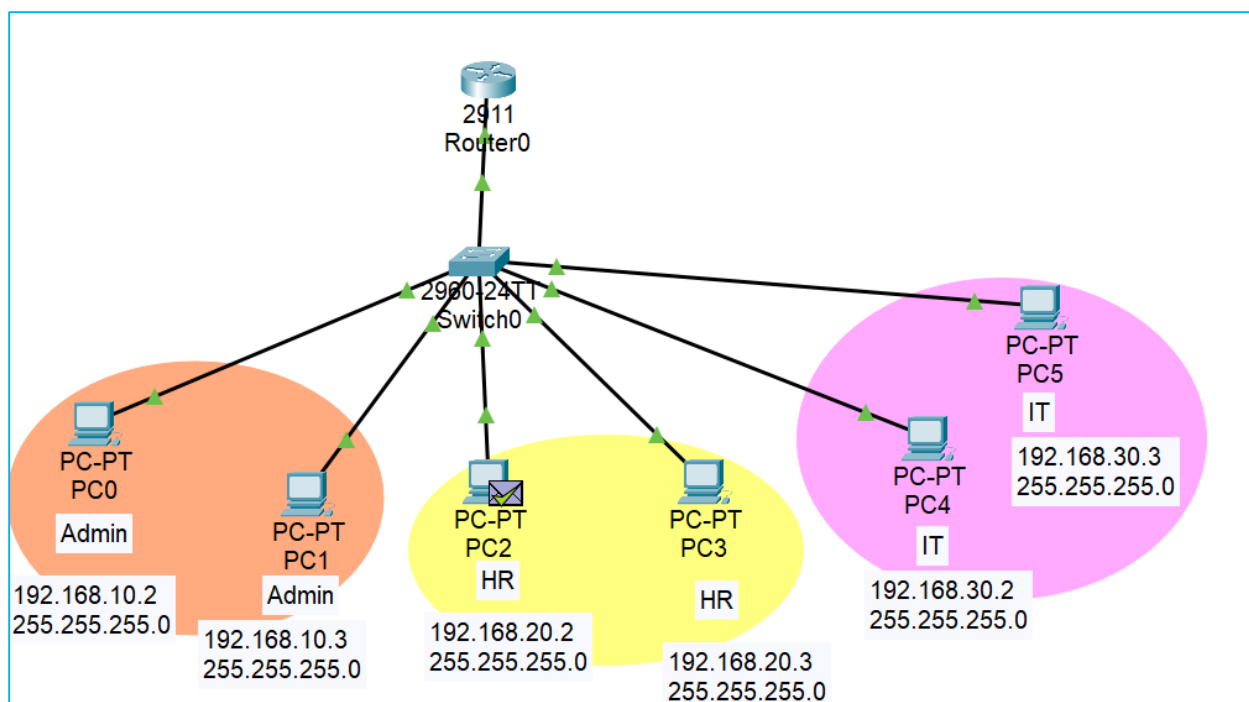
| In Layers                                                                                                     | Out Layers |
|---------------------------------------------------------------------------------------------------------------|------------|
| Layer7                                                                                                        | Layer7     |
| Layer6                                                                                                        | Layer6     |
| Layer5                                                                                                        | Layer5     |
| Layer4                                                                                                        | Layer4     |
| Layer 3: IP Header Src. IP: 192.168.20.1, Dest. IP: 192.168.20.2<br>ICMP Message Type: 3 ICMP Message Type: 8 | Layer3     |
| Layer 2: Ethernet II Header<br>000B.BE20.E101 >> 0002.17C3.84B4                                               | Layer2     |
| Layer 1: Port FastEthernet0                                                                                   | Layer1     |

1. The packet's destination IP address matches the device's IP address or the broadcast address. The device de-encapsulates the packet.
2. The packet is an ICMP packet. The ICMP process processes it.
3. The ICMP process received an Administratively Prohibited Unreachable message.
4. The Ping process received an Echo Reply message.



**Fig 6: Ping from HR to Admin – Simulation Result**





**Fig 7: Ping from HR to IT – Simulation Result**

| Event List |           |      |
|------------|-----------|------|
|            | At Device | Type |
|            | PC2       | ICMP |
|            | Switch0   | ICMP |
|            | Router0   | ICMP |
|            | Switch0   | ICMP |
|            | PC4       | ICMP |
|            | PC5       | ICMP |
|            | Switch0   | ICMP |
|            | Router0   | ICMP |
|            | Switch0   | ICMP |
|            | PC2       | ICMP |

## 12. Packet Tracer File



Secure Enterprise Network.pkt